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Detecting cancer lesions in mammographic images using machine learning techniques

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Detecting cancer lesions in mammographic images using machine learning

Abstract. Breast cancer remains a leading cause of mortality among women, necessitating accurate early detection via mammography. This thesis presents a Deep Learning-based Computer-Aided Diagnosis (CAD) system designed to assist radiologists in classifying breast abnormalities. Utilizing the MIAS dataset, the study employs a Transfer Learning approach with the ResNet50V2 architecture to distinguish between benign and malignant pathologies. The engineering pipeline integrates Contrast Limited Adaptive Histogram Equalization (CLAHE) and geometric augmentation to address challenges related to data scarcity and low image contrast.

A comparative analysis was conducted between Global Classification (whole-image) and Region-Based (ROI) Classification. The Global model yielded suboptimal results with a classification accuracy of 34 percent, attributing failure to the loss of fine textural details during downsampling. Conversely, the Region-Based model, which focuses spatial resolution on the lesion, achieved a classification accuracy of 83 percent and a malignant sensitivity of 87 percent. These findings demonstrate that while global approaches struggle with small datasets, region-aware Deep Learning models offer high diagnostic sensitivity, validating their potential as effective clinical decision support tools.

Keywords: Breast Cancer, Deep Learning, Mammography, Convolutional Neural Networks, Transfer Learning, ResNet50, MIAS Dataset, Medical Image Analysis.

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1. Introduction

Breast cancer remains a significant global health challenge and is currently the most prevalent cancer diagnosed in women worldwide. Early detection is the most critical factor in improving survival rates and treatment outcomes. While mammography serves as the gold standard for screening, the accurate interpretation of mammographic images is a complex task prone to human error. This thesis explores the application of Deep Learning, specifically Convolutional Neural Networks (CNNs), to assist in the automated detection and classification of breast tumors.

1.1. Background and Motivation

Breast cancer accounts for a substantial percentage of cancer-related mortality globally. According to international health statistics, regular screening programs are essential for identifying abnormalities before clinical symptoms appear. Mammography, an X-ray imaging technique, is the primary modality used for these screenings.

However, the interpretation of mammograms relies heavily on the expertise of radiologists. This manual process is subject to several limitations:

- Fatigue and Volume: Radiologists often review hundreds of images daily, leading to visual fatigue and a potential decrease in diagnostic accuracy over time.
- Inter-observer Variability: Different experts may interpret the same image differently, leading to inconsistent diagnoses.
- False Positives and Negatives: Dense breast tissue can mask small tumors (false negatives), or normal tissue overlapping can mimic malignancies (false positives), leading to unnecessary biopsies or missed diagnoses.

These challenges motivate the development of Computer-Aided Diagnosis (CAD) systems. By leveraging Artificial Intelligence (AI), specifically Deep Learning, it is possible to provide radiologists with a "second opinion," thereby reducing oversight errors and improving diagnostic consistency.

1.2. Problem Statement

Diagnosing breast cancer from mammograms is inherently difficult due to the visual nature of the medical images. Unlike natural images (e.g., distinguishing a cat from a dog), mammograms are low-contrast, grayscale images where the region of interest (the tumor) often shares a similar pixel intensity distribution with the surrounding healthy fibro-glandular tissue.

The specific challenges addressed in this project include:

1. Low Contrast and Noise: Pathological features such as micro-calcifications or architectural distortions can be subtle and easily obscured by image noise or dense tissue.

2. Data Scarcity: Training Deep Learning models typically requires massive datasets. In the medical domain, annotated datasets are scarce due to privacy concerns and the high cost of expert labeling.
3. Class Imbalance: In screening datasets, the number of healthy patients vastly outnumbers those with pathology. A model trained on such data risks becoming biased toward the majority class (healthy), failing to detect the minority class (cancer).

1.3. Project Objectives

The primary objective of this thesis is to develop a robust Deep Learning pipeline for the detection and classification of breast abnormalities using the MIAS (Mammographic Image Analysis Society) dataset. This dataset categorizes abnormalities into distinct classes such as Calcifications (CALC), Circular Masses (CIRC), Spiculated Masses (SPIC), and Asymmetries (ASYM).

To address the aforementioned challenges, this research focuses on the following specific goals:

- Development of a Transfer Learning Pipeline: To utilize the ResNet50V2 architecture, pre-trained on the ImageNet dataset, to overcome the limitations of the small dataset size (322 images).
- Data Preprocessing and Augmentation: To implement advanced preprocessing techniques, including Contrast Limited Adaptive Histogram Equalization (CLAHE), to enhance tissue structures. Furthermore, geometric augmentation is applied to artificially expand the training set and mitigate overfitting.
- Comparative Experimental Analysis: To evaluate and compare two distinct classification strategies:
  1. Experiment 1 (Global Classification): Classifying the entire mammogram image. This approach tests the model's ability to automatically locate features within the full scan.
  2. Experiment 2 (Region-Based Classification): Classifying extracted Regions of Interest (ROIs). This approach utilizes ground-truth coordinates provided in the dataset metadata to focus the neural network specifically on the lesion texture, simulating a diagnostic support tool where the radiologist indicates a suspicious area.
- Handling Class Imbalance: To implement cost-sensitive learning strategies, such as weighted loss functions, to ensure the model prioritizes the detection of malignant cases over the majority healthy class.

  The complete source code, including preprocessing pipelines and model training scripts, is available at: `https://github.com/kwiatkowskamarta/Mammography_cancer_detection.git`

2. Theoretical Background and Literature Review

This chapter establishes the comprehensive theoretical framework necessary to understand the methodologies applied in this thesis. The discussion is structured into three primary domains: the medical context of breast cancer diagnosis via mammography, the fundamental principles of Digital Image Processing and Deep Learning, and a critical review of existing literature regarding Computer-Aided Diagnosis (CAD) systems.

2.1. Medical Background: Breast Cancer and Mammography

Breast cancer is a malignant proliferation of epithelial cells lining the ducts or lobules of the breast. It represents a major global health burden, being the most frequently diagnosed cancer in women. From an anatomical perspective, the breast is a complex structure composed of lobules (milk-producing glands), ducts (tubules transporting milk to the nipple), and the stroma (fatty and connective tissue surrounding the ducts and lobules, blood vessels, and lymphatic vessels).

2.1.1. Mammography Screening Principles

Mammography remains the gold standard modality for early breast cancer detection. It utilizes low-energy X-rays (typically generated by molybdenum or rhodium anodes) to create distinct images of the internal breast structure. The fundamental principle relies on the differential attenuation of X-rays by various tissue types.

- Fatty tissue appears radiolucent (dark gray/black).
- Fibro-glandular tissue appears radiopaque (white/light gray).
- Calcifications and masses appear highly radiopaque (bright white).

This contrast mechanism, however, presents a significant diagnostic challenge in patients with dense breasts, where the abundance of glandular tissue can mask underlying abnormalities, a phenomenon known as the "masking effect."

2.1.2. Taxonomy of Mammographic Abnormalities

The precise classification of abnormalities is crucial for developing robust Machine Learning models. The MIAS (Mammographic Image Analysis Society) dataset utilized in this research adheres to standard radiological definitions.

1. Circumscribed Masses (CIRC): These are space-occupying lesions with well-defined, distinct margins. While a sharp margin often suggests a benign etiology (e.g., cyst or fibroadenoma), malignancy cannot be ruled out without biopsy, especially in older patients.
2. Spiculated Masses (SPIC): These lesions are characterized by radiating spicules or "spikes" extending from a central tumor mass into the surrounding tissue. This stellate appearance is highly pathognomonic for malignancy (Invasive Ductal Carcinoma) and represents a high-priority target for CAD systems.

3. Calcifications (CALC): These appear as small, bright deposits of calcium. Macro-calcifications are typically benign, whereas micro-calcifications - appearing as clusters of tiny specks - are often the earliest sign of Ductal Carcinoma In Situ (DCIS). Detecting these requires high spatial resolution, which poses a challenge for downsampled neural networks.
4. Architectural Distortion (ARCH): In these cases, the normal breast architecture is distorted with no definite mass visible. Features may include focal retraction or tethering of the parenchyma.
5. Asymmetry (ASYM): This refers to a density deposit that is visible in one breast but lacks a corresponding counterpart in the other.
6. Miscellaneous (MISC): Ill-defined findings that do not strictly fit the morphological criteria of the above categories but represent a deviation from normal tissue structure.

Each abnormality is further stratified by clinical severity: Benign (B) or Malignant (M). Additionally, healthy scans are labeled as Normal (NORM), indicating the absence of any suspicious findings.

2.2. Digital Image Processing Techniques

Raw mammograms often exhibit suboptimal contrast, particularly in the dense tissue regions where tumors reside. Preprocessing is therefore an essential step to enhance feature visibility for the neural network.

2.2.1. Histogram Equalization limitations

Standard Histogram Equalization (HE) works by redistributing pixel intensities to achieve a uniform histogram. While this increases global contrast, it often leads to two specific artifacts in mammography:

1. Over-enhancement of background noise (the dark air region).
2. Loss of detail in areas of similar density, "washing out" subtle tumor boundaries.

2.2.2. Contrast Limited Adaptive Histogram Equalization (CLAHE)

To address the limitations of global HE, this study employs CLAHE. Unlike global methods, CLAHE operates on small, localized regions of the image called "tiles" (in this study, an 8×8 grid was selected).

The algorithm proceeds in three steps:

1. Tiling: The image is divided into non-overlapping contextual regions.
2. Histogram Clipping: In each tile, the histogram is computed. To prevent noise amplification, the histogram is "clipped" at a predefined threshold (Clip Limit β). Pixels above this threshold are redistributed equally among all bins.
3. Bilinear Interpolation: To prevent blocking artifacts at tile boundaries, the neighboring tiles are combined using bilinear interpolation.

This technique effectively enhances local tissue texture - such as the edges of a mass or individual calcifications - without saturating the entire image.

2.3. Deep Learning Fundamentals

Deep Learning constitutes a subset of Machine Learning methods based on Artificial Neural Networks (ANNs) with representation learning. Unlike traditional algorithms that require manual feature extraction (e.g., Haralick textures), Deep Learning models automatically learn hierarchical feature representations from raw data.

2.3.1. Convolutional Neural Networks (CNNs)

Convolutional Neural Networks are the cornerstone of modern medical image analysis. While fully connected Multi-Layer Perceptrons (MLPs) fail to capture spatial dependencies and require a prohibitive number of parameters for high-resolution images, CNNs introduce parameter sharing and translation invariance.

A typical CNN architecture consists of alternating convolutional and pooling layers, followed by fully connected layers.

2.3.2. The Convolution Operation

The core mechanism of a CNN is the convolution of the input image I with a learnable kernel (filter) K. For a 2D image, the feature map S at position (i, j) is calculated as:

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i - m, j - n) \cdot K(m, n) \tag{1}$$

During training, the network updates the weights in kernel K to detect specific features, such as vertical edges, color gradients, or complex textures like spicules.

2.3.3. Non-Linear Activation (ReLU)

Linear operations alone cannot model complex biological data. To introduce non-linearity, an activation function is applied element-wise. The Rectified Linear Unit (ReLU) is defined as:

$$f(x) = \max(0, x) \tag{2}$$

ReLU is preferred over Sigmoid or Tanh functions because it mitigates the "vanishing gradient" problem, allowing for the training of deeper networks, and induces sparsity in the activations.

2.3.4. Pooling Layers

Pooling layers reduce the spatial dimensions of the feature maps, thereby decreasing the computational load and controlling overfitting.

- Max Pooling: Selects the maximum value in a window.

- Global Average Pooling (GAP): Calculates the mean of the entire feature map. In this project, GAP is utilized in the final classification head to reduce the feature tensor to a vector, minimizing the number of trainable parameters.

2.4. Transfer Learning and Architecture

Training deep CNNs from scratch requires massive datasets to converge. In medical imaging, where annotated data is scarce (e.g., the MIAS dataset contains only 322 images), training a deep network from random initialization typically results in severe overfitting.

2.4.1. Transfer Learning Strategy

Transfer Learning leverages the knowledge gained from solving one problem and applies it to a different but related problem. In this thesis, we utilize a network pre-trained on ImageNet (a database of 1.2 million natural images). The initial layers of the network act as generic feature extractors (detecting edges, curves), which are universally applicable to visual data, including mammograms.

2.4.2. ResNet50V2 Architecture

We employ the ResNet50V2 architecture. Standard deep networks suffer from the degradation problem: as network depth increases, accuracy saturates and then degrades. Residual Networks (ResNet) address this by introducing "skip connections."

2.4.3. The Residual Block

Instead of learning a direct mapping $H(x)$, a residual block learns a residual function $F(x) = H(x) - x$. The original input x is added to the output of the stacked layers:

$$y = F(x, \{W_i\}) + x \tag{3}$$

This identity shortcut allows gradients to flow through the network unimpeded during backpropagation.

2.4.4. Version 2 (Pre-activation)

The V2 variant of ResNet, utilized in this study, applies Batch Normalization and ReLU activation *before* the convolution operation. This "pre-activation" structure improves the regularization of the model and has been shown to offer superior performance on transfer learning tasks compared to the original ResNetV1.

2.5. Literature Review

The evolution of CAD systems for mammography has progressed from handcrafted feature engineering to end-to-end Deep Learning.

2.5.1. Traditional Approaches

Early methodologies relied on extracting textural descriptors. Techniques such as Gray Level Co-occurrence Matrices (GLCM), Gabor filters, and Local Binary Patterns (LBP) were commonly used to quantify tissue heterogeneity. These features were then fed into classifiers like Support Vector Machines (SVM) or Random Forests. While effective on small datasets, these methods lack robustness against variations in scanner types and biological noise.

2.5.2. Deep Learning Approaches

The advent of CNNs revolutionized the field.

- Patch-based Classifiers: Works by Chougrad et al. (2018) demonstrated high accuracy using InceptionV3 on small image patches. However, these approaches often assume the lesion location is already known.
- Whole-image Classifiers: Shen et al. (2019) developed networks capable of analyzing full mammograms. Their research highlighted the difficulty of detecting small lesions in high-resolution scans without significant downsampling, which results in information loss.

2.5.3. Identified Research Gap

A significant gap remains in effectively utilizing small datasets like MIAS for automated detection without massive overfitting. Many existing studies prioritize accuracy on balanced subsets, neglecting the realistic class imbalance and the "vanishing tumor" problem inherent in resizing full images. This thesis addresses this gap by explicitly comparing a global resizing strategy against a semi-automated Region-Based (ROI) strategy, quantifying the trade-off between automation and diagnostic sensitivity.

3. Methodology

This chapter details the engineering pipeline developed to detect and classify breast abnormalities in mammograms. The methodology is structured into four main phases: data acquisition and cleaning, image preprocessing and augmentation, neural network architecture design, and the experimental framework comparing global versus region-based classification strategies.

The entire pipeline was implemented using the Python programming language (v3.10+), utilizing the TensorFlow/Keras framework for Deep Learning and OpenCV for computer vision operations.

3.1. Dataset Description and Preparation

The study utilizes the Mammographic Image Analysis Society (MIAS) database. The dataset consists of 322 digitized medio-lateral oblique (MLO) views. Each image is associated with ground-truth metadata provided in a text file (`info.txt`), which details the character of the background tissue, the class of abnormality, the severity, and the spatial coordinates of the lesion.

3.1.1. Metadata Parsing and Cleaning

Raw medical data is rarely ready for direct ingestion by machine learning algorithms. Based on the structure of the source files, a custom parsing algorithm was developed to standardize the dataset labels. The following cleaning steps were implemented:

1. Handling Normal Cases: The raw metadata represents healthy cases with the label 'NORM' and missing values for severity and coordinates. The parsing logic imputes these missing values, assigning a severity label of 'Normal' and setting spatial coordinates (x, y, r) to zero.
2. Duplicate Management: Several images in the MIAS dataset contain multiple findings (e.g., a calcification and a mass in the same breast). To formulate a distinct classification problem, the dataset was consolidated to one entry per image. A severity hierarchy was established:

$$Malignant > Benign > Normal \tag{4}$$

   In cases of multiple annotations for a single `REFNUM`, the pathology with the highest severity score was retained. This ensures the model is trained to prioritize the most critical clinical diagnosis.

3.1.2. Class Imbalance Analysis

An initial statistical analysis of the dataset revealed a significant class imbalance. The majority of scans are Normal ($N \approx 200$), while Malignant cases represent the minority ($N < 60$).

To prevent the neural network from converging on a trivial solution (predicting the majority class for all inputs), Class Weighting was calculated and applied during the training phase (detailed in Section 3.6).

3.2. Image Preprocessing Pipeline

Raw mammograms are high-resolution grayscale images often characterized by low contrast and significant noise. A uniform preprocessing pipeline was applied to all images prior to feeding them into the neural network.

3.2.1. Contrast Enhancement (CLAHE)

Mammographic lesions often exhibit pixel intensities very similar to dense fibro-glandular tissue, making them difficult to detect. To address this, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied.

Unlike global histogram equalization, which stretches the contrast of the entire image and can over-amplify noise in background regions, CLAHE operates on small localized regions called "tiles".

- Clip Limit: Set to 2.0 to limit the amplification of noise in uniform regions.
- Grid Size: Set to (8×8) tiles.

This transformation effectively enhances the definition of edges and texture irregularities, which are critical features for identifying spiculated masses and architectural distortions.

3.2.2. Resizing and Normalization

The original scans (1024×1024 pixels) were downsampled to 224×224 pixels. This dimension was selected to match the input tensor requirements of the pre-trained ResNet50V2 architecture. Pixel intensity values were normalized to the range $[0, 1]$ by dividing by 255, ensuring numerical stability during gradient descent optimization.

3.3. Data Splitting and Augmentation

3.3.1. Patient-Level Splitting strategy

A critical requirement in medical imaging is the prevention of *Data Leakage*. Standard random splitting could result in an augmented version of an image appearing in the training set while the original appears in the test set, leading to artificially inflated accuracy.

To ensure the validity of the results, data splitting was performed at the Patient Level using the unique Reference ID (`REFNUM`).

- Training Set: 80% of unique patients.
- Test Set: 20% of unique patients.

Stratified sampling was utilized to ensure that the distribution of Malignant, Benign, and Normal cases remained consistent across both subsets.

3.3.2. Geometric Augmentation

Given the small size of the dataset ($N = 322$), deep models are prone to overfitting. To artificially expand the training distribution, geometric augmentation was applied only to the training set. The transformations included:

- Horizontal Flip: Mirroring the image left-to-right.
- Vertical Flip: Mirroring the image top-to-bottom.

This effectively tripled the size of the training corpus, providing the network with more varied examples of lesion orientation without altering the pathological features.

3.4. Experimental Design

To evaluate the impact of spatial attention on diagnostic accuracy, two distinct experimental protocols were designed and implemented.

3.4.1. Experiment 1: Global Classification (Baseline)

Objective: To assess the feasibility of fully automated diagnosis where the model analyzes the entire mammogram.

- Input: Full breast images resized to 224×224.
- Classes: 3 (Normal, Benign, Malignant).
- Hypothesis: The model may struggle to detect small lesions due to the loss of high-frequency detail during downsampling.

3.4.2. Experiment 2: Region-Based (ROI) Classification

Objective: To simulate a "Second Opinion" diagnostic tool where the model analyzes specific regions identified as suspicious.

- Input: Cropped Regions of Interest (ROIs).
- Extraction Logic: Using the ground-truth coordinates (x, y, r) provided in the metadata, a square patch was extracted centered on the abnormality. The patch size was calculated as $1.2 \times r$ to include surrounding contextual tissue.
- Classes: 2 (Benign vs. Malignant).
- Hypothesis: By removing the background noise and focusing resolution on the lesion texture, sensitivity to malignancy will improve.

3.5. Deep Learning Architecture

The core classification engine is based on Transfer Learning. We utilized ResNet50V2 (Residual Network with 50 layers, Version 2) pre-trained on the ImageNet database.

3.5.1. Feature Extractor

The convolutional base of ResNet50V2 was instantiated with pre-trained weights. These layers were frozen (non-trainable) during the training phase. This allows the model to leverage robust, low-level feature detectors (edges, curves, textures) learned from millions of generic images, applying them to the mammography domain.

3.5.2. Custom Classification Head

The original fully connected layers of ResNet were removed and replaced with a custom classification head designed for this specific task. The architecture of the head is as follows:

1. Global Average Pooling 2D: Reduces the spatial dimensions of the feature maps ($7 \times 7 \times 2048$) to a single feature vector (1×2048).
2. Dense Layer: A fully connected layer with 128 neurons and ReLU activation. This layer learns non-linear combinations of the extracted features specific to mammography.
3. Dropout: A regularization layer with a rate of 0.5. It randomly deactivates 50% of neurons during each training step to prevent co-adaptation and overfitting.
4. Output Layer: A Dense layer with Softmax activation, outputting the probability distribution over the target classes.

3.6. Training Configuration

The models were trained using the Adam optimizer with a learning rate of $\eta = 10^{-4}$. The loss function employed was Categorical Cross-Entropy:

$$L = -\sum_{i=1}^{C} y_i \cdot \log(\hat{y}_i) \tag{5}$$

where C is the number of classes, y is the true label (one-hot encoded), and $\hat{y}$ is the predicted probability.

3.6.1. Handling Class Imbalance

To address the imbalance described in Section 3.1, cost-sensitive learning was implemented. Class weights w_j were calculated inversely proportional to class frequencies:

$$w_j = \frac{N_{total}}{N_{classes} \times N_j} \tag{6}$$

This ensures that the penalty for misclassifying a minority class (Malignant) is significantly higher than for the majority class (Normal/Benign), forcing the optimizer to prioritize cancer detection.

3.6.2. Regularization and Callbacks

Two Keras callbacks were utilized to ensure optimal convergence and prevent overfitting:

- Early Stopping: Training is halted if the Validation Loss does not improve for a set number of epochs (Patience = 8), preventing the model from memorizing noise.
- Model Checkpoint: Only the model weights yielding the lowest Validation Loss were saved, ensuring the best generalizing model is used for the final evaluation.

4. Experimental Results

This chapter presents the quantitative and qualitative results obtained from the Deep Learning pipeline. The performance of the system is evaluated using standard statistical metrics including Accuracy, Precision, Recall, and the Area Under the Receiver Operating Characteristic Curve. The analysis is divided into two distinct sections corresponding to the experimental design: the baseline performance of the Global Classification model and the optimized performance of the Region-Based model.

4.1. Evaluation Metrics

To objectively assess the predictive capabilities of the models, performance metrics were calculated based on the confusion matrix elements: True Positives, True Negatives, False Positives, and False Negatives.

Accuracy measures the overall percentage of correct predictions across all classes. While useful as a general indicator, it can be misleading in imbalanced datasets where the majority class dominates.

Precision quantifies the proportion of positive identifications that were actually correct. In this study, it indicates how many of the predicted malignant cases were truly cancerous.

Recall, often referred to as Sensitivity in medical literature, measures the proportion of actual positive cases that were correctly identified by the model. In the context of cancer screening, this is the most critical metric. A low recall implies a high rate of false negatives, meaning cancerous lesions are missed, which can lead to delayed treatment and poor patient outcomes.

The F1-Score provides the harmonic mean of Precision and Recall, offering a single metric that balances the trade-off between the two, which is particularly useful when the class distribution is uneven.

4.2. Experiment 1: Global Classification Analysis

In the first experiment, the ResNet50V2 model was trained on full mammogram images resized to 224 by 224 pixels. The objective was to classify cases into three categories: Normal, Benign, and Malignant.

4.2.1. Quantitative Performance

The global classification model demonstrated limited predictive capability, achieving an overall test accuracy of approximately 34 percent. The detailed classification metrics reveal a significant disparity in performance across classes. The model achieved moderate recall for the Normal class but failed to generalize to the pathological classes. Specifically, the recall for Malignant cases was 0.00, indicating that the model did not correctly identify any cancerous lesions in the test set.

Class	Precision	Recall	F1-Score	Support
Normal	0.60	0.43	0.50	42
Benign	0.13	0.33	0.19	12
Malignant	0.00	0.00	0.00	10

Table 4.1. Classification Report for Experiment 1 using full mammogram images.

4.2.2. Failure Analysis

The confusion matrix for this experiment illustrates that the model predictions were heavily biased towards the majority class or distributed randomly among the pathological classes without discerning the underlying features.

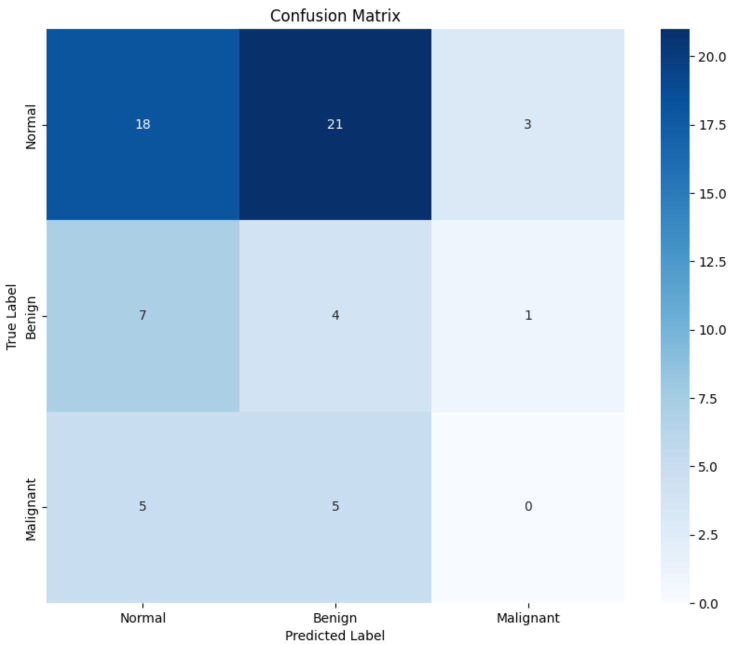

Figure 4.1. Confusion Matrix for Global Classification showing high misclassification rates.

The inability of the global model to detect malignancy can be attributed to the low signal-to-noise ratio inherent in the experimental design. By resizing the entire breast anatomy to a low resolution, the spatial features of small lesions, such as micro-calcifications or spicules, were effectively lost. The neural network likely focused on dominant low-frequency features such as the pectoral muscle or the overall density of the breast, which are not reliable indicators of malignancy.

4.3. Experiment 2: Region-Based Classification Analysis

In the second experiment, the methodology shifted to a focused Region of Interest approach. The model was trained on cropped image patches containing the lesion to distinguish between Benign and Malignant pathologies.

4.3.1. Quantitative Performance

The region-based model demonstrated a substantial improvement in diagnostic performance, achieving an overall test accuracy of 83 percent. Unlike the global model, this approach successfully learned to differentiate between the textural properties of benign and malignant masses.

Most notably, the recall for Malignant cases reached 0.87. This indicates that the system correctly flagged 87 percent of the cancerous lesions present in the test set. The precision for the Malignant class was 0.85, suggesting a low rate of false positives.

Class	Precision	Recall	F1-Score	Support
Benign	0.79	0.78	0.78	40
Malignant	0.85	0.87	0.86	60

Table 4.2. Classification Report for Experiment 2 using Region of Interest patches.

4.3.2. Confusion Matrix and ROC Analysis

The confusion matrix for the region-based experiment confirms a strong diagonal correlation, representing correct predictions for both classes.

To further validate the discriminative ability of the model, the Receiver Operating Characteristic curve was generated. The Area Under the Curve metric serves as a robust aggregate measure of performance across all possible classification thresholds. The high AUC value observed in this experiment confirms that the model assigns higher probability scores to malignant cases compared to benign ones with high consistency.

4.4. Comparative Summary

A direct comparison of the two experimental protocols highlights the impact of input resolution on model performance. The global classification approach yielded a malignant recall of 0 percent, rendering it unsuitable for diagnostic support. In contrast, the region-based approach achieved a malignant recall of 87 percent.

This disparity confirms that the convolutional neural network requires high-resolution textural information to identify the subtle morphological features associated with breast cancer. When the background noise is removed and the resolution is focused on the lesion, the ResNet50V2 architecture is highly effective at feature extraction in the mammography domain.

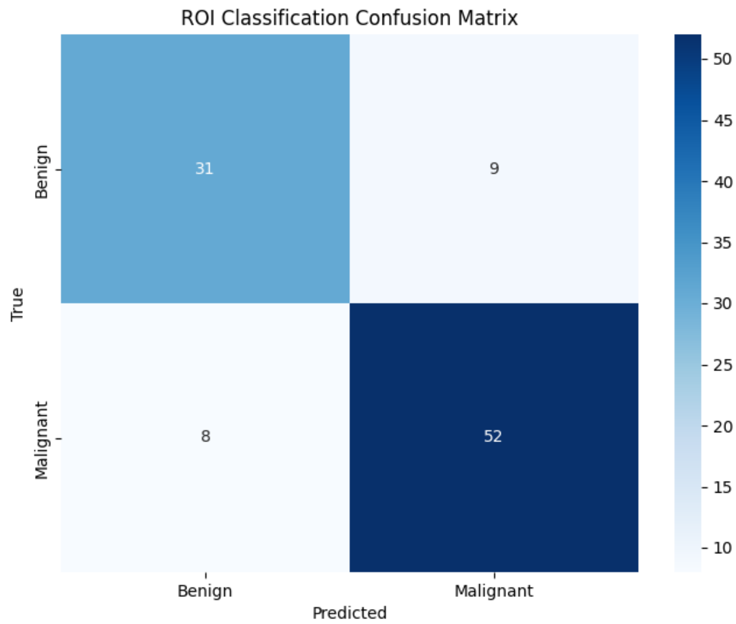

Figure 4.2. Confusion Matrix for ROI Classification demonstrating improved sensitivity.

5. Discussion

This chapter provides a critical interpretation of the experimental findings presented in the previous chapter. It analyzes the underlying factors contributing to the performance disparity between the global and region-based classification models, discusses the clinical relevance of the achieved sensitivity, evaluates the impact of the implemented engineering techniques, and acknowledges the limitations inherent in the current study design.

5.1. Analysis of Classification Strategies

The most significant finding of this research is the pronounced difference in diagnostic capability between the global classification approach and the region-based approach. While the global model failed to identify malignant cases entirely, the region-based model achieved a high degree of sensitivity. This disparity can be explained by examining the relationship between image resolution and feature preservation.

5.1.1. The Resolution Trade-off and Feature Loss

In the global classification experiment, the input mammograms were downsampled from their original high resolution to 224 by 224 pixels to accommodate the architecture of the ResNet50V2 model. A typical breast tumor in the MIAS dataset may occupy a physical region corresponding to a small fraction of the total breast area. When the entire image is aggressively downsampled, the spatial resolution available to represent the tumor is drastically reduced.

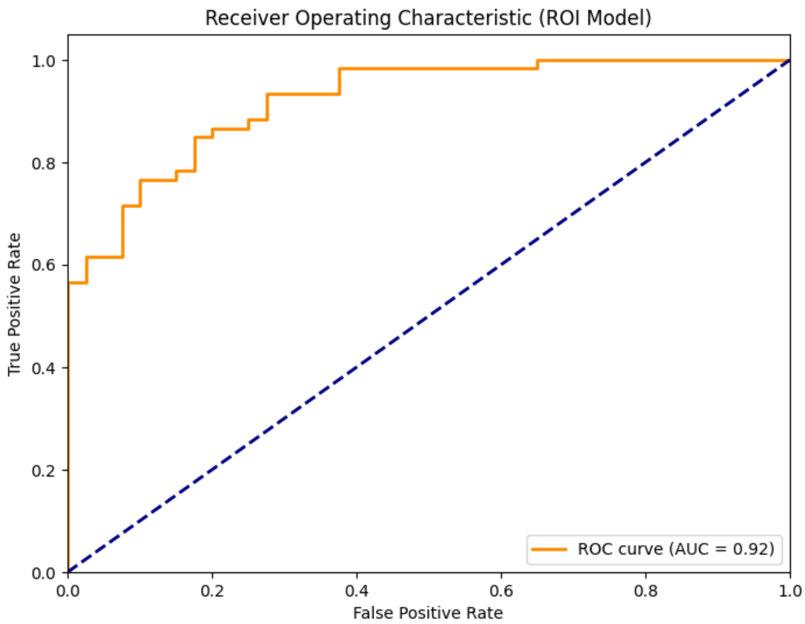

Figure 4.3. Receiver Operating Characteristic curve for the ROI model.

Pathological features critical for malignancy detection, such as the fine, radiating spicules associated with invasive ductal carcinoma or the granular texture of micro-calcifications, rely on high-frequency spatial information. In the global model, these fine details are effectively smoothed out during resizing, becoming indistinguishable from the background noise and normal glandular tissue. Consequently, the neural network lacks the necessary discriminative information to form a decision boundary between benign and malignant classes, resulting in the convergence towards the majority class.

5.1.2. Signal-to-Noise Ratio in ROI Processing

The region-based experiment circumvented the resolution issue by cropping the image around the coordinate of the abnormality. By feeding the network a patch where the lesion occupies a significant portion of the visual field, the signal-to-noise ratio is substantially improved. In this context, the signal represents the pathological texture, while the noise represents the surrounding healthy tissue and background air.

Because the Region of Interest patch is also resized to 224 by 224 pixels, the pixel density per unit of lesion area remains high. This allows the convolutional filters within the deep learning model to successfully extract texture-based features, such as edge roughness and density heterogeneity, which correlate strongly with malignancy. This confirms that for current convolutional architectures, preserving local resolution is more critical than maintaining global context when the target object is small relative to the image size.

5.2. Clinical Relevance and Sensitivity

The primary metric for evaluating any medical screening system is sensitivity, or recall. In the context of breast cancer screening, the cost of a false negative error is high, as it represents a missed diagnosis that delays treatment and worsens patient prognosis. Conversely, a false positive error, while undesirable due to patient anxiety and the cost of follow-up procedures, is considered less critical than missing a cancer.

The region-based model achieved a recall of 0.87 for malignant cases. This indicates that the system correctly flagged 87 percent of the cancerous lesions presented in the test set. While this does not reach the perfection required for a standalone diagnostic tool, it falls within a range that makes it viable as a decision support system. A radiologist using such a system could receive a probabilistic assessment of a suspicious region, potentially alerting them to subtle features of malignancy that might otherwise be overlooked due to visual fatigue.

5.3. Impact of Preprocessing and Augmentation

The results also highlight the importance of the auxiliary engineering steps in the pipeline, particularly given the constraints of the digitized analog dataset used.

5.3.1. Contrast Enhancement

Mammographic images are characterized by low contrast, where the pixel intensity differences between a mass and the surrounding dense tissue are minimal. The application of Contrast Limited Adaptive Histogram Equalization proved effective in mitigating this issue. By enhancing local contrast without over-amplifying noise in uniform regions, this preprocessing step emphasized the boundaries of masses. This likely contributed to the model's ability to distinguish between the circumscribed margins typical of benign masses and the indistinct or spiculated margins typical of malignant masses.

5.3.2. Data Augmentation and Generalization

Deep learning models require large datasets to generalize effectively. Training a complex architecture like ResNet50V2 on the limited number of samples available in the MIAS dataset poses a high risk of overfitting, where the model memorizes the training data but fails on unseen test data. The implementation of geometric data augmentation, specifically horizontal and vertical flipping, artificially increased the diversity of the training set. This forced the network to learn features that are invariant to orientation, leading to a more robust model that generalized well to the test set, as evidenced by the convergence of training and validation metrics.

5.4. Limitations of the Study

Despite the promising results obtained from the region-based model, several limitations must be acknowledged when interpreting these findings.

5.4.1. Dataset Constraints

The study utilized the MIAS dataset, which consists of digitized analog mammograms. Modern clinical practice relies on Full-Field Digital Mammography, which offers significantly higher dynamic range and resolution. The artifacts and noise inherent in digitized film may negatively impact model performance compared to native digital images. Furthermore, the dataset size is relatively small compared to modern standards, limiting the statistical power of the evaluation.

5.4.2. Semi-Automated Workflow

The high performance of the region-based model is predicated on the availability of accurate lesion coordinates. In this study, ground-truth annotations were used to extract the regions of interest. In a real-world clinical setting, this workflow is semi-automated, requiring either a radiologist or a separate detection algorithm to first localize the suspicious area. The model currently functions as a classifier of known lesions rather than a fully automated detector of occult disease.

5.4.3. Interpretability

The neural network operates as a black box, providing a probability score without explicit justification. In healthcare applications, explainability is crucial for building trust with clinicians. The current implementation does not produce visual heatmaps or localization maps that would indicate which specific pixels or features influenced the decision to classify a lesion as malignant.

6. Conclusion and Future Work

This thesis presented the design, implementation, and evaluation of a deep learning framework for the automated detection of breast cancer in mammographic images. By leveraging the ResNet50V2 architecture and transfer learning, the study investigated the efficacy of convolutional neural networks in distinguishing between benign and malignant abnormalities within the MIAS dataset.

6.1. Conclusion

The primary objective of this research was to determine the feasibility of using computer vision techniques to assist radiologists in the interpretation of mammograms, a task complicated by the low contrast of medical X-rays and the subtle nature of pathological features. Through a comparative experimental design, the study highlighted critical engineering constraints and identified effective strategies for overcoming them.

The investigation demonstrated that a global classification approach, which resizes the entire mammogram to a standard input dimension, is insufficient for reliable diagnosis on this specific dataset. The loss of high-frequency spatial details during downsampling prevented the model from capturing the minute textural indicators of malignancy, resulting in a system that failed to identify cancerous cases.

However, the transition to a region-based classification strategy yielded significant improvements. By isolating the region of interest and maintaining a high pixel density over the lesion, the model achieved a classification accuracy of 83 percent and a malignant recall of 87 percent. This result confirms that the ResNet50V2 architecture is capable of extracting discriminative features from mammographic tissue when the input data is preprocessed correctly to preserve local resolution.

The integration of contrast limited adaptive histogram equalization and geometric data augmentation proved to be essential components of the pipeline. These techniques successfully mitigated the challenges of low image contrast and data scarcity, enabling the training of a robust classifier despite the limited size of the dataset. The final system stands as a proof of concept for a decision support tool that can provide a probabilistic second opinion on suspicious regions identified by medical professionals.

6.2. Future Work

While the results of the region-based model are promising, the transition from an academic prototype to a clinical tool requires further development. Future research should focus on automation, data scale, and model interpretability.

6.2.1. Fully Automated Detection

The current successful model relies on semi-automated input, requiring pre-defined coordinates to crop the tumor. The next logical step in this research trajectory is the implementation of object detection architectures, such as You Only Look Once (YOLO) or Faster R-CNN. These models are designed to regress bounding box coordinates and classify objects simultaneously. Integrating such a network would allow the system to ingest a full mammogram, automatically locate suspicious regions, and classify them without human intervention, effectively bridging the gap between the two experiments conducted in this thesis.

6.2.2. Dataset Expansion

The validity of the current findings is constrained by the small size and analog nature of the MIAS dataset. To improve the robustness and clinical applicability of the model, training should be extended to modern, digital mammography datasets such as INbreast or the CBIS-DDSM collection. These datasets offer higher resolution, greater dynamic range, and a significantly larger number of samples, which would allow for the training of deeper architectures and potentially improve generalization to unseen patient populations.

6.2.3. Explainable Artificial Intelligence

A significant barrier to the adoption of deep learning in medicine is the black-box nature of neural networks. Future iterations of this project should incorporate explainable AI techniques. Methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) can generate visual heatmaps overlaid on the original mammogram. These maps would highlight the specific pixels and textures that influenced the model's decision, providing radiologists with visual evidence to support or refute the automated diagnosis and increasing trust in the system.

6.2.4. Multiclass Severity Grading

The current study simplified the problem into a binary classification of benign versus malignant. Future work could expand the scope to include multiclass classification that distinguishes between specific types of abnormalities, such as calcifications versus masses, or predicts the BI-RADS assessment category. This would provide more granular diagnostic information that aligns better with standard radiological reporting protocols.

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